1. Project Overview and Objectives

The goal of this project is to conduct a rigorous analysis of **Brent crude oil price dynamics** from May 1987 to September 2022 using **Bayesian change point detection techniques**. The primary objective is to identify **structural breaks** in the time series and analyze how major geopolitical events—such as wars, sanctions, political decisions, and OPEC policy changes—correlate with these shifts.

By understanding when and how significant events impact oil markets, the analysis aims to deliver **data-driven insights** that can support:

- Investment strategy development
- Policy decision-making
- Operational planning for stakeholders in the global energy sector

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2. Data Analysis Workflow & Phased Approach

The analysis follows a phased, structured workflow designed to ensure methodological rigor and actionable outcomes.

Phase 1: Data Acquisition & Preparation (Current Focus – Task 1)

- Collect historical Brent oil prices (May 20, 1987 – September 30, 2022)
- Clean data, address missing values, and ensure date parsing integrity
- Compile a structured dataset of 10–15 significant geopolitical and economic events with approximate start dates for contextual analysis

Phase 2: Exploratory Data Analysis (EDA) & Feature Engineering

- Investigate time series properties: trend, stationarity, seasonality
- Transform event data into model-ready features (e.g., binary indicators, lagged effects)
- Visualize historical price behavior alongside major global events

Phase 3: Model Development – Bayesian Change Point Detection

- Implement Bayesian change point models using **PyMC3**
- Identify statistically significant shifts in price behavior (e.g., mean or variance changes)

Phase 4: Model Validation & Interpretation

- Validate model outputs using MCMC diagnostics and visual inspection
- Align detected change points with relevant historical events
- Interpret the magnitude, direction, and persistence of identified structural changes

Phase 5: Insight Generation & Communication

- Translate analytical results into clear insights
- Deliver findings via formal reports, visualizations, and an **interactive dashboard** tailored for analysts, policymakers, and industry experts

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3. Detailed Data Preparation Strategy

Initial Data Ingestion

- Parse 'Date' and 'Price' columns properly
- Convert the date to `datetime` format and set it as the time index
- Verify data completeness and check for missing or abnormal values

Event Dataset Compilation

- Identify major geopolitical and economic events (e.g., Gulf Wars, 9/11, COVID-19 pandemic, Russia-Ukraine war, major OPEC decisions)
- Capture key metadata: event name, type, approximate date, and short description
- Structure data for integration into the model as exogenous indicators

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4. Modeling and Interpretation Framework

The analysis applies **Bayesian change point models** to the time series data to:

- Detect **structural breaks** in the mean or variance of Brent oil prices
- Estimate **posterior distributions** of change points
- Compare breakpoints with historical events to assess correlation

Expected Outputs

- A list of estimated change points (with confidence intervals)
- Visualization of structural shifts in the time series
- Interpretation of each shift in the context of contemporaneous events
- Quantitative measures of event-driven impact duration and volatility change

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5. Assumptions and Limitations

Core Assumptions

- **Data Integrity**: Brent price data is assumed accurate and complete
- **Event Relevance**: Selected geopolitical/economic events are assumed impactful enough to influence price movements
- **Model Appropriateness**: Bayesian change point detection is suitable for this analysis context

Limitations and Caveats

It is important to note that **temporal correlation does not imply causation**. While structural breaks may coincide with historical events, the oil market is influenced by a **multitude of overlapping factors** (e.g., demand shocks, global recessions, speculative trading). For instance:

After the 9/11 attacks, oil prices initially spiked due to uncertainty but later declined sharply due to global demand concerns—demonstrating the **nonlinear** and **multifactorial** nature of price dynamics.

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6. Proposed GitHub Repository Structure

```
brent-oil-change-point-analysis/
│
├── data/
│   ├── brent_prices.csv
│   └── events_dataset.csv
│
├── notebooks/
│   ├── 01_data_preparation.ipynb
│   ├── 02_eda_and_feature_engineering.ipynb
│   ├── 03_modeling_change_points.ipynb
│   └── 04_event_correlation_analysis.ipynb
│
├── models/
│   └── pymc3_models/
│
├── reports/
│   └── interim_report.pdf
│   └── final_report.pdf
│
├── dashboard/
│   └── app/
│       ├── backend/
│       └── frontend/
│
└── README.md
```

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7. Communication and Stakeholder Engagement

To ensure accessibility and stakeholder alignment:

- **Interactive Dashboard**: An intuitive web dashboard will enable users to explore price trends, detected change points, and event overlays interactively
- **Formal Reports**: Structured deliverables will be provided at each phase—highlighting key insights, model decisions, and policy implications